How Empirically Normal Are Sample Means? Testing the CLT on UP School Teachers

1. Team Members

Student ID	Contribution
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Week / Assignment no.	8
Dataset used	UDISE Dataset of Uttar Pradesh
Software used	Python, Colab
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2. Objective of the Report

2.1 Objective

The objective of the report is to illustrate what proportion of sample means of the sampling distribution of means, are above 1, 2 and 3 standard deviations more than the mean

2.2 Question, hypothesis, or case

Table 1. Hypotheses or cases examined in this report.

Label	A question or a hypothesis, stated with a direction	What result would show us we were wrong
H1	The empirical proportions of sample means beyond +1 SD, +2 SD, and +3 SD differ from the corresponding proportions expected under a normal distribution.	The statistics, if matched with those of the normal distribution, result in ~15.87% of values above mean + 1 standard deviation, ~2.28% above 2 standard deviations, and 0.135% above 3 standard deviations.

2.3 Justification

The sampling distribution of sample means from repeated samples resembles a normal distribution, provided the sample size is sufficiently large. This allows us to represent that the proportions are at par with the standard normal distribution, with ~15.87% of values above mean + 1 standard deviation, ~2.28% above 2 standard deviations, and 0.135% above 3 standard deviations.

3. Data Used and Analysis Description

3.1 The dataset

The dataset used was the UDISE dataset of Uttar Pradesh.

3.2 Variables selected

Table 2. Variables used in this analysis.

Variable (as named in the data)	What it measures, and in what units	Measurement scale	Role	Why selected
total_tch	Measures the count of total teachers in a school	Discrete (Quantitative)	Variable used for sampling distributio n	To evaluate the total number of teachers in a school

Variables considered and left out

Other UDISE variables were excluded because they were categorical and therefore not suitable for this analysis.

3.3 Data cleaning

As the variable was directly available in the School Profile dataset, no cleaning was performed.

Table 3. Size of the working dataset

Cases (rows)	Variables (columns)
264965	1 (total_tch)

3.4 Statistical methods and why

Table 4. Methods used, by hypothesis or case.

Label	Method used	Why is this method appropriate here	Alternative rejected, and why
H1	Repeated	Repeatedly drawing 10,000 random samples of fixed size (500) and calculating their means allows us to construct the sampling distribution of the sample mean and estimate its standard deviation, as well as the proportions of sample means above μ + 1SD, μ + 2SD, and μ + 3SD.	**Single random sample**, because one sample provides only one sample mean and cannot empirically demonstrate the distribution of sample means or estimate the proportions across repeated samples.

3.4 Assumptions Checked

Table 5. Assumptions Checked.

Assumption	How we checked it	Does it hold?	If not, what we did
Independent observations	Each sample was randomly drawn with replacement from the school-level dataset, so observations across draws were treated as independent.	Yes	-
Consistent sampling procedure	Used the same random-sampling procedure for each sample size.	Yes	-

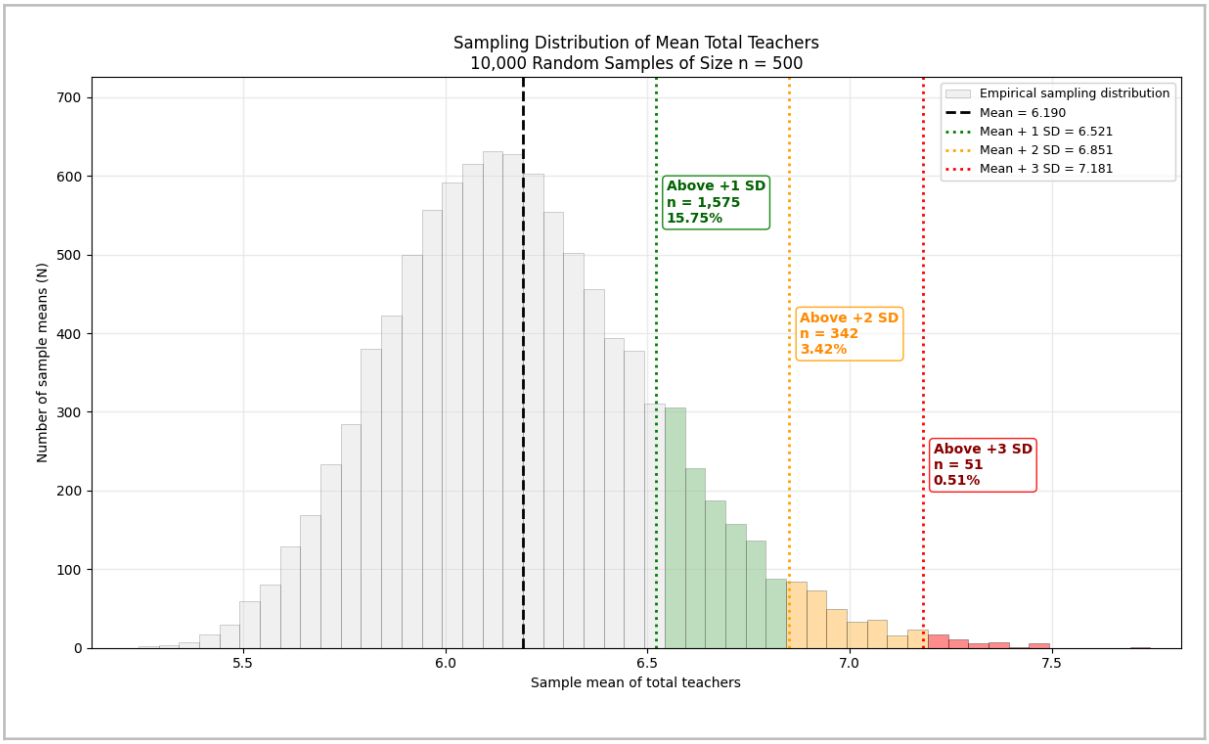
4. Analysis Findings

4.1 Proportion of Sample means above +1 S.D., +2 S.D., +3 S.D.

Table 6. Findings for proportion of sample means above +1 S.D., +2 S.D., +3 S.D.

Threshold	Percentage Above	Normal Expected (%)	Difference (percentage points)
μ	46.630%	50.000%	-3.370
μ + 1 SD	15.750%	15.866%	-0.116
μ + 2 SD	3.420%	2.275%	+1.145
μ + 3 SD	0.510%	0.135%	+0.375

Figure 1

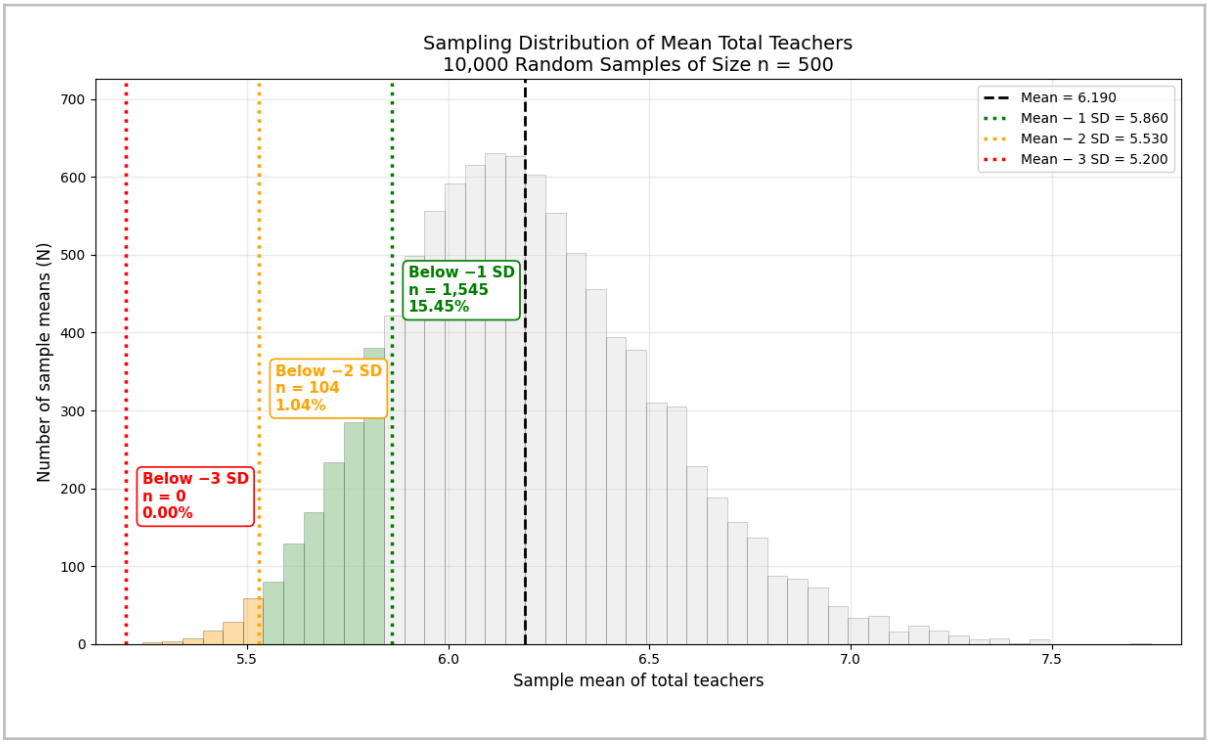


Caption. Proportion of values above μ + 1 S. D., μ + 2 S. D., and μ + 3 S. D.

What we read from this figure: Above μ + 1 S. D, 15.75% (compared to 15.866% of normal curve) of the sample means lie, 3.42% (compared to 2.275% of normal curve) lie for μ + 2 S. D, and 0.510% (compared to 0.135% of normal curve) lie for μ + 3 S. D.

4.2 Proportion of Sample means below -1 S.D., -2 S.D., -3 S.D.

Figure 2

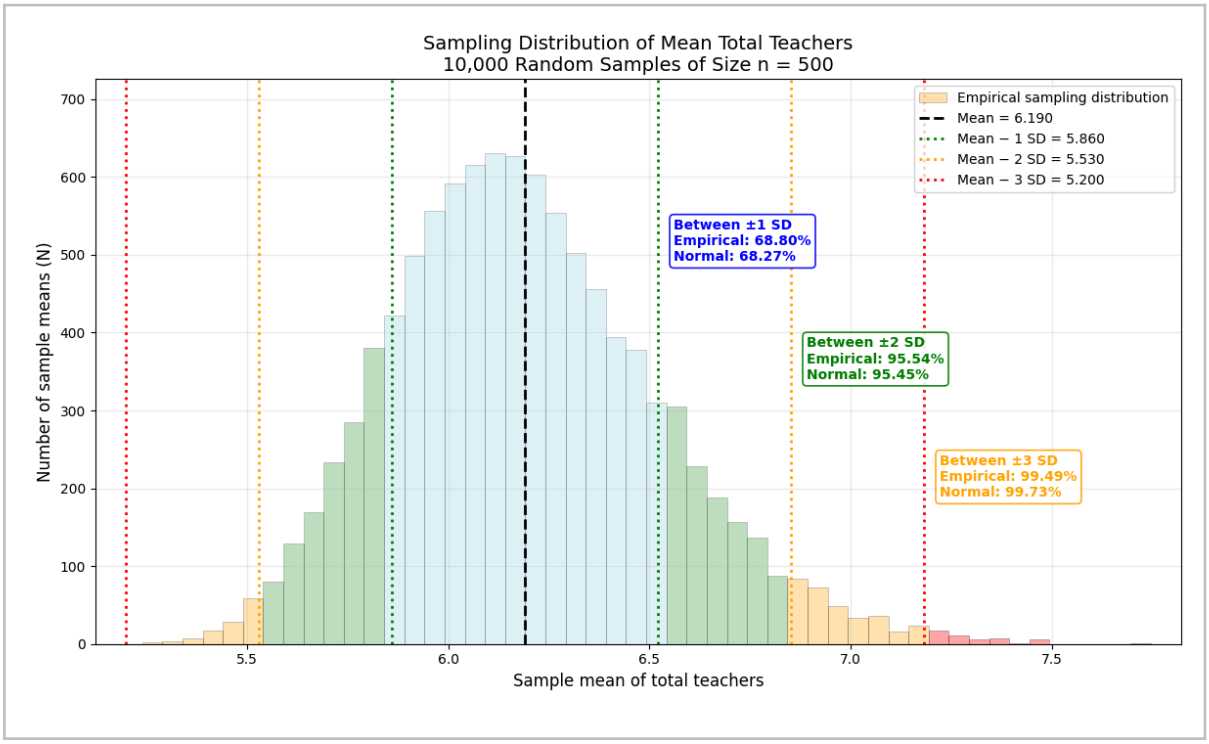


Caption. Proportion of values below μ - 1 S. D., μ - 2 S. D., and μ - 3 S. D.

What we read from this figure: Below μ - 1 S. D, 15.45% (compared to 15.866% of normal curve) of the sample means lie, 1.04% (compared to 2.275% of the normal curve) lie for μ - 2 S. D, and 0% (compared to 0.135% of the normal curve) lie for μ - 3 S. D.

4.3 Proportion of Sample means between ± 1 S.D., ± 2 S.D., ± 3 S.D.

Figure 3

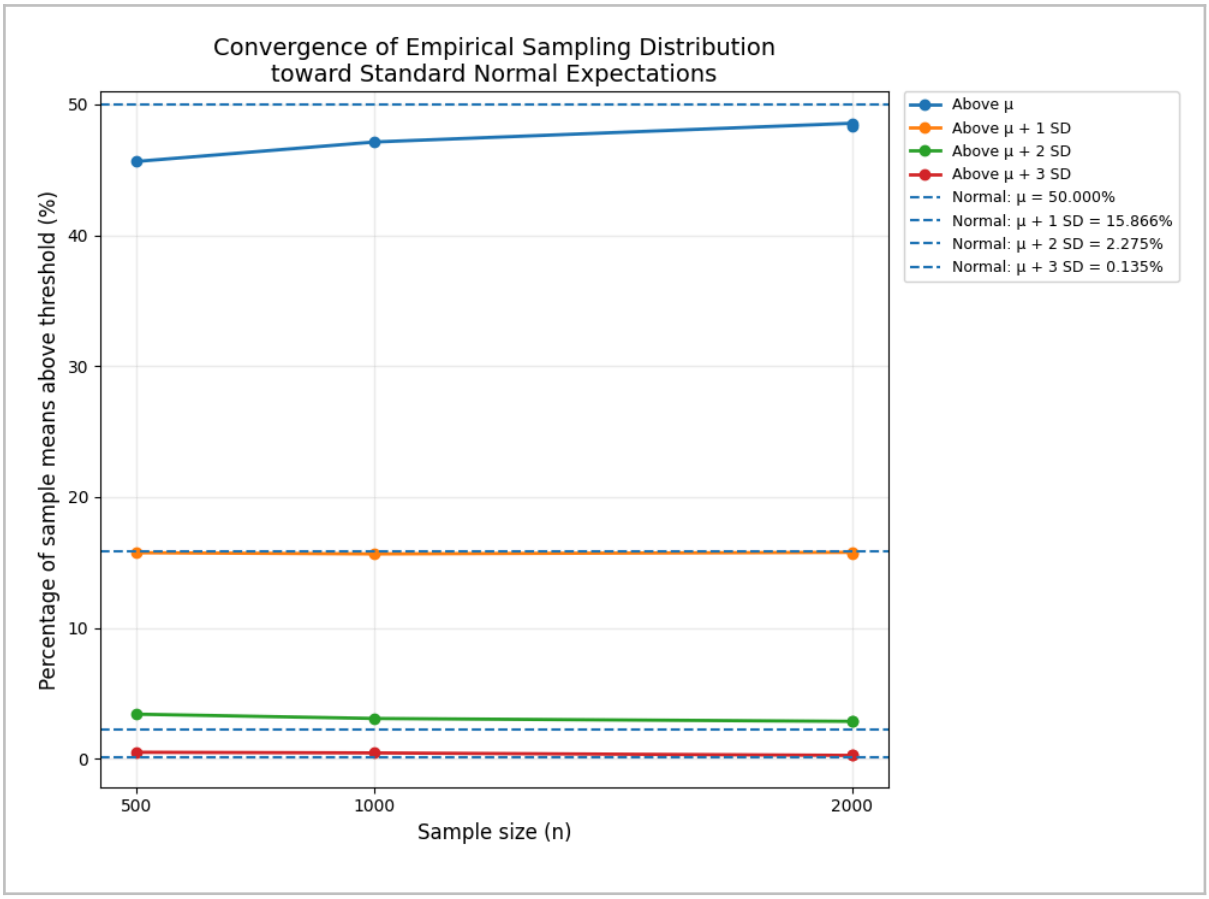


Caption. Proportion of values between μ ± 1 S. D., μ ± 2 S. D., and μ ± 3 S. D.

What we read from this figure: Between μ ± 1 S. D, 68.8% (compared to 68.27% of normal curve) of the sample means lie, 95.54% (compared to 95.45% of normal curve) lie for between μ ± 2 S. D, and 99.49% (compared to 99.7% of the normal curve) lie for μ ± 3 S. D.

4.4 Effect of sample size on closeness to the normal curve

Figure 4



Caption. Proportion of values closer to normal curve on increasing sample size

What we read from this figure: As we increase sample size, proportion of values above 1 S.D., 2 S.D. and 3 S.D. tend to match the normal curve as the distribution also approaches normal distribution.

Table 7. Comparison of convergence towards normal distribution across sample sizes.

Threshold	Percentage Above (n = 500)	Percentage Above (n = 1000)	Percentage Above (n = 2000)	Normal Expected (%)
μ	46.630%	47.120%	48.550%	50.000%
μ + 1 SD	15.750%	15.660%	15.780%	15.866%
μ + 2 SD	3.420%	3.090%	2.870%	2.275%
μ + 3 SD	0.510%	0.460%	0.270%	0.135%

What we read from this table: As the sample size increases, we can see convergence of proportions towards normal distribution.

Results :

Verdict on the hypothesis

☐ Supported ☐ Partly supported ☑ Not supported

5. Key Highlights for the Class, and Insights

5.1 Four things worth the class's attention

Table 8. Highlights from the Report

#	Highlight	Why it matters to the rest of the class
1	The 68-95-99.7% rule is mostly followed, especially at the centre	Due to Central Limit Theorem, the sampling distribution of means follows a normal curve, and approaches population statistics, and therefore allows us to draw inferences from a single sufficiently large sample.
2	The middle matches the normal curve well. At n = 500, 15.75% of sample means lie above μ + 1 SD (expected 15.87%), and 68.8% lie within ±1 SD (expected 68.27%)	Teacher counts are skewed, yet the sampling distribution still looks normal near the center. That's the Central Limit Theorem working
3	The tails don't match as well. At n = 500, 3.42% lie above μ + 2 SD (expected 2.275%) and 0.51% above μ + 3 SD (expected 0.135%).	The right tail is slightly stretched by schools with very high teacher counts, so normal rules of thumb can misjudge extremes.
4	Bigger n helps, but slowly. Above μ + 2 SD fell from 3.42% (n = 500) to 2.87% (n = 2000).	"Large enough n" depends on the data. Skewed data need a bigger n.

5.2 What we learned

Sampling distributions tend to follow normal distribution faster in the middle and slowly in the tails. It is because, in the middle, in balanced samples, highs and lows cancel out, leading to a bell shape. However, at tails, It takes a very large sample size to dilute the impact of an extreme outlier. Therefore, the tails take much longer to converge to normal distribution after the middle looks perfect. Coverage at ±1, ±2 and ±3 SD (68.8%, 95.54%, 99.49%) stayed close to the 68-95-99.7 rule. But only 46.63% of sample means were above μ, not 50%, which shows a slight asymmetry that shrinks as n grows.

5.3 What this analysis cannot tell us

- It covers one variable (total_tch) from one state, so the convergence speed may differ elsewhere.
- Only three sample sizes were tested, so we can't say where the tails fully settle.
- We compared by eye and ran no formal normality test, so "close enough" is a judgement call.

5.4 What we would do next

We would try more sample sizes (30, 100, 300, 5000), add a Q-Q plot to check if the dataset follows normal distribution (appears as a straight line in QQ plot) and skewness measure, and repeat the analysis on another variable like total students to compare convergence towards the normal curve.

Appendix. Code

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https://colab.research.google.com/drive/1GzSfCu7u0Em8km2auexU-86AFu_PtHCc?usp=sharing